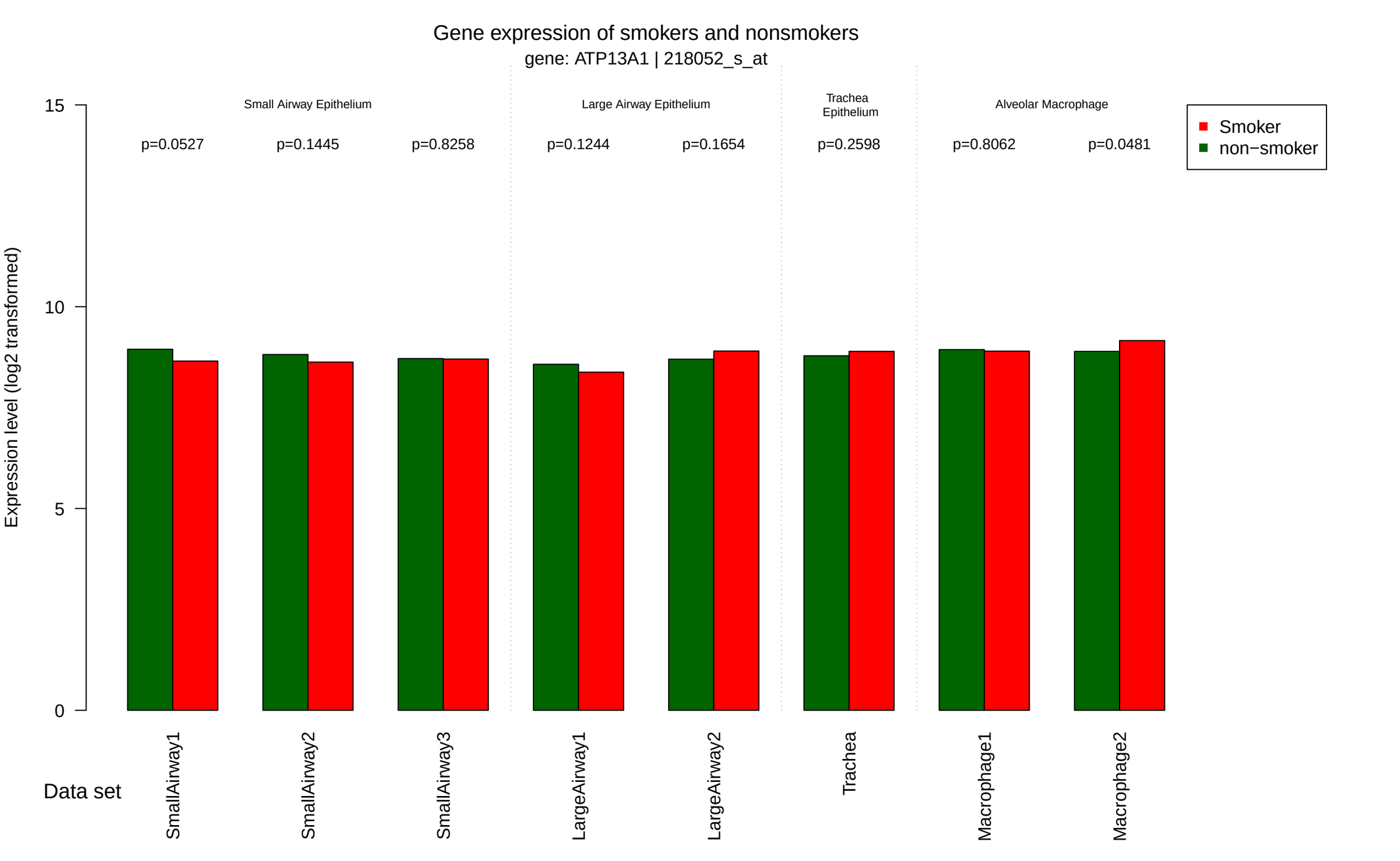
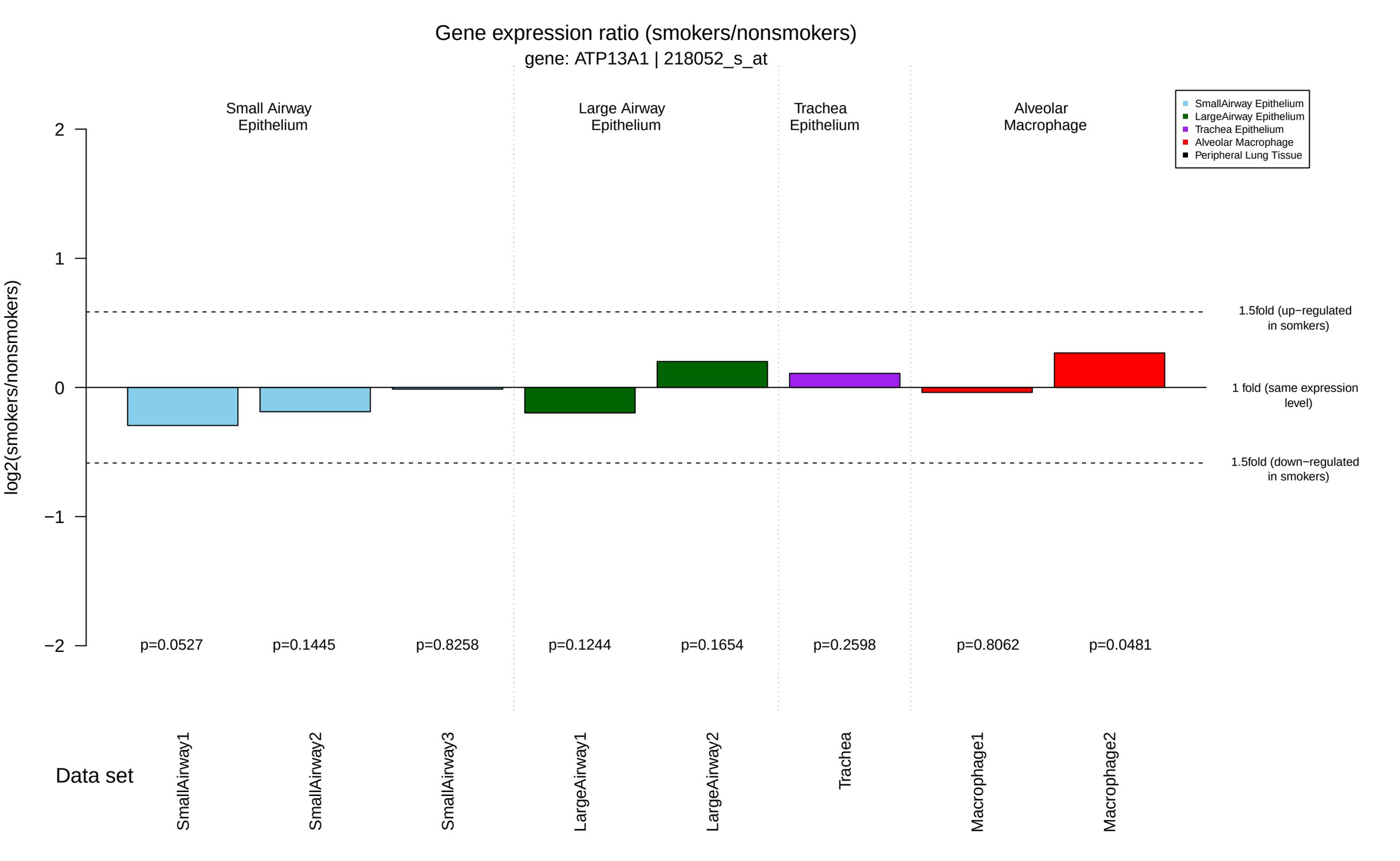
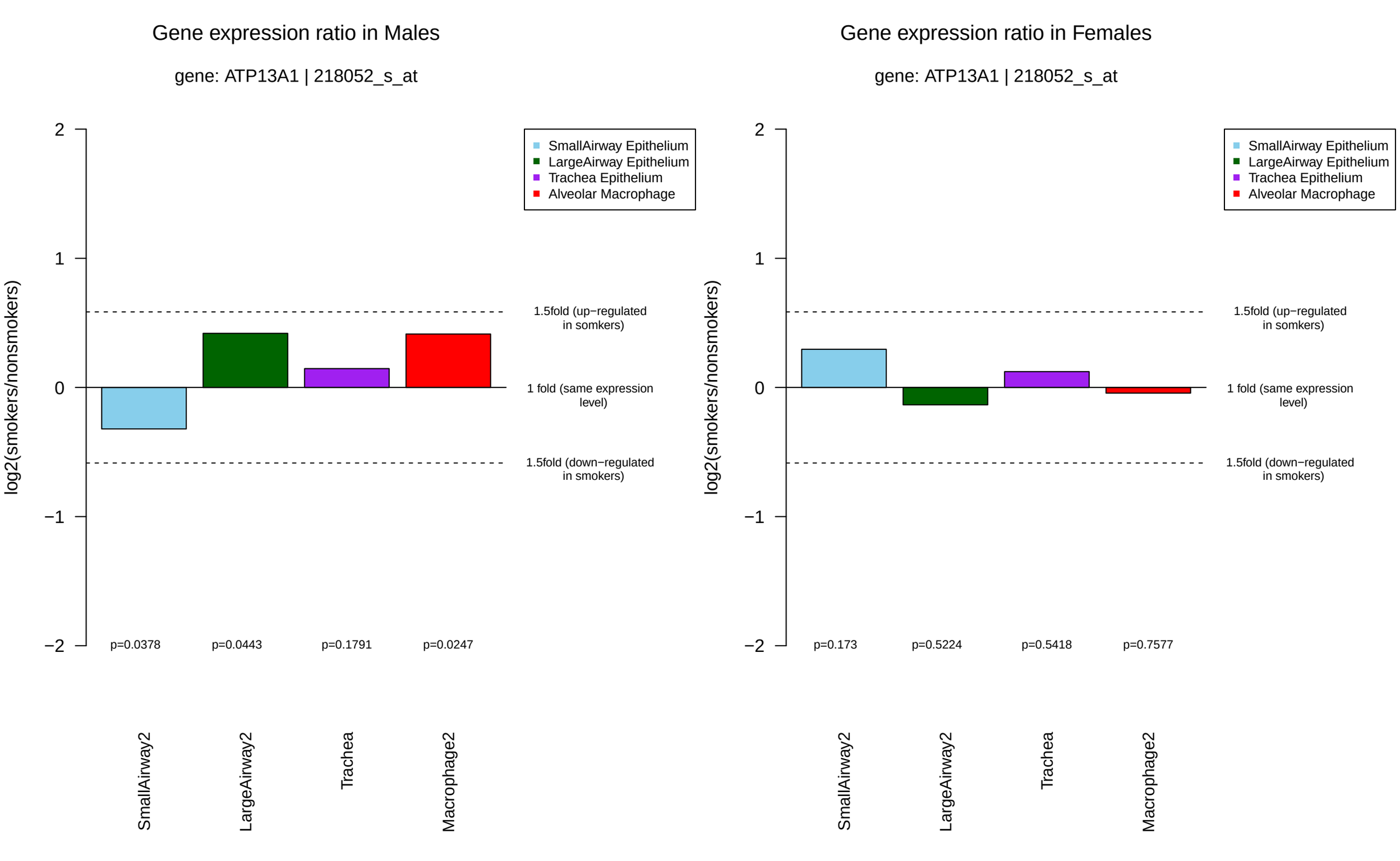
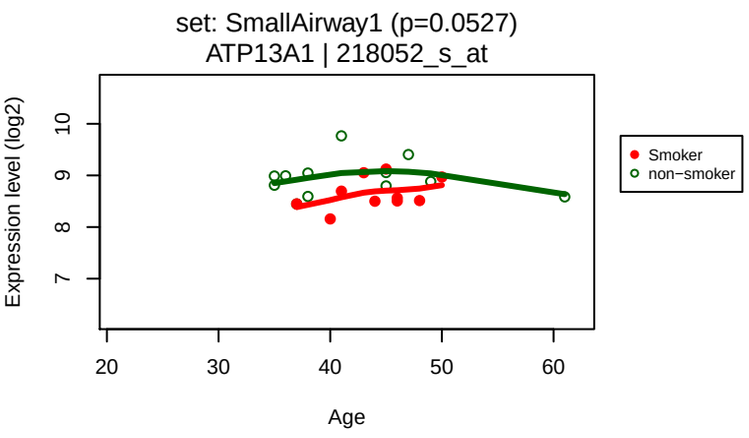
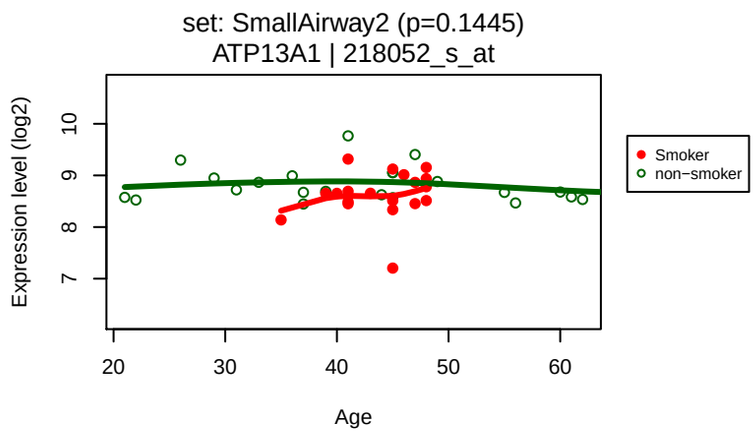
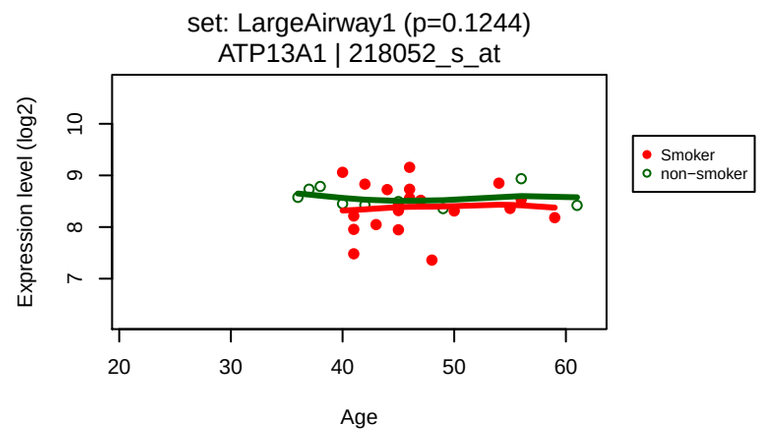

Gene expression of smokers and nonsmokers
gene: ATP13A1 | 218052_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0527
p=0.1445
p=0.8258
p=0.1244
p=0.1654
p=0.2598
p=0.8062
p=0.0481
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

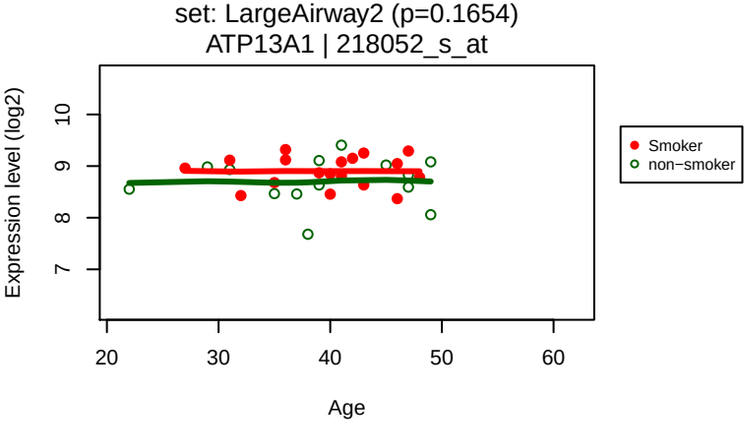

Gene expression ratio (smokers/nonsmokers)
gene: ATP13A1 | 218052_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0527
p=0.1445
p=0.8258
p=0.1244
p=0.1654
p=0.2598
p=0.8062
p=0.0481
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

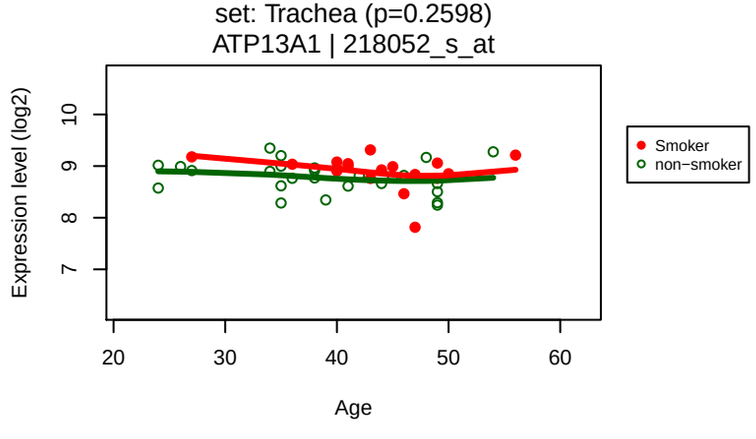

Gene expression ratio in Males
gene: ATP13A1 | 218052_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0378
p=0.0443
p=0.1791
p=0.0247
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP13A1 | 218052_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.173
p=0.5224
p=0.5418
p=0.7577
SmallAirway2
LargeAirway2
Trachea
Macrophage2

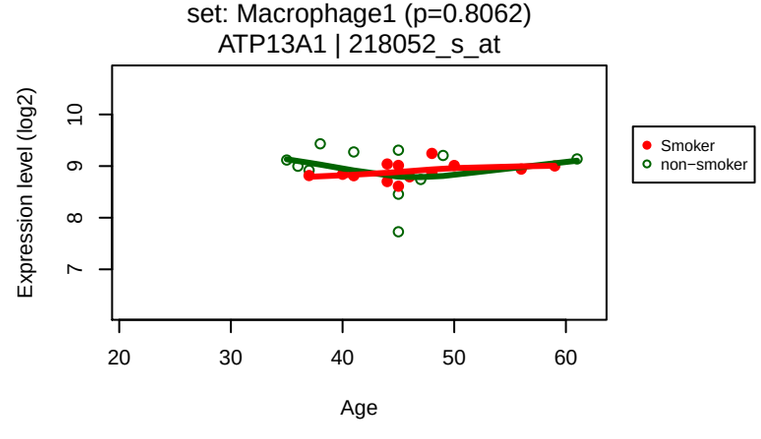

set: SmallAirway1 (p=0.0527)
ATP13A1 | 218052_s_at
Expression level (log2)
Age
Smoker
non−smoker

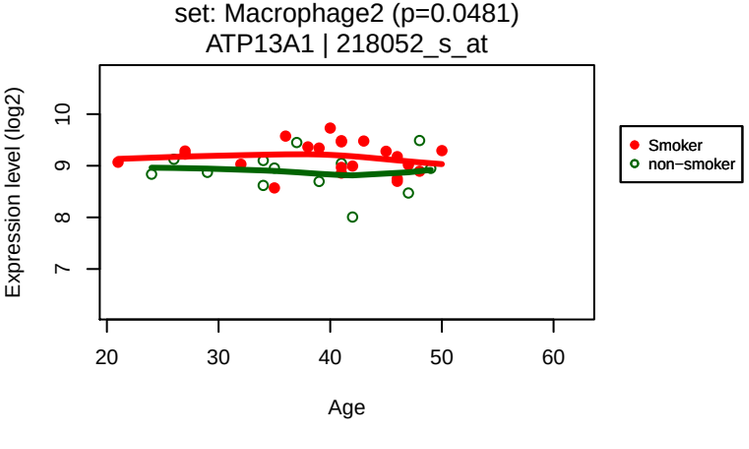

set: SmallAirway2 (p=0.1445)
ATP13A1 | 218052_s_at
Expression level (log2)
Age
Smoker
non−smoker

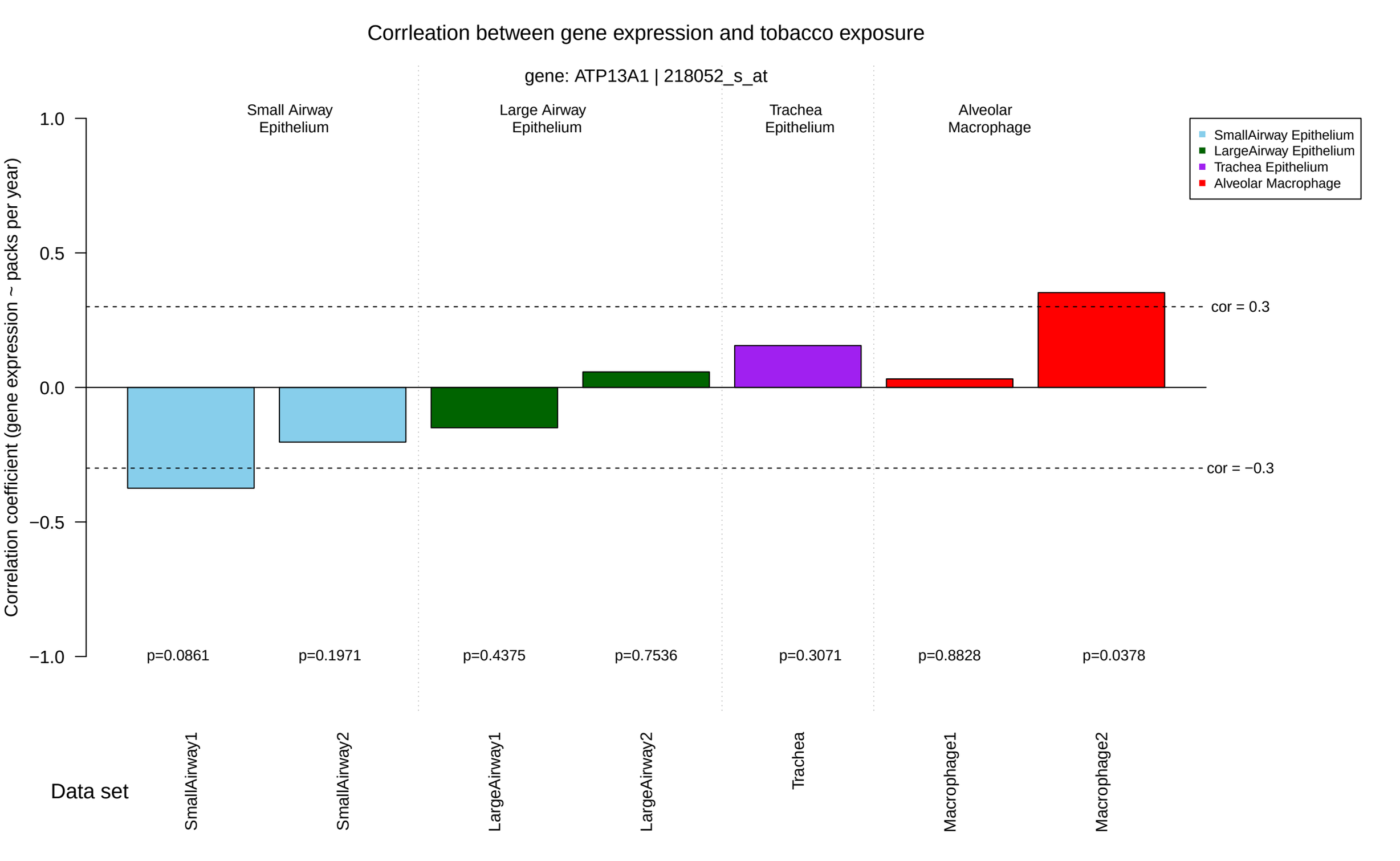

set: LargeAirway1 (p=0.1244)
ATP13A1 | 218052_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1654)
ATP13A1 | 218052_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2598)
ATP13A1 | 218052_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8062)
ATP13A1 | 218052_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0481)
ATP13A1 | 218052_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP13A1 | 218052_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0861
p=0.1971
p=0.4375
p=0.7536
p=0.3071
p=0.8828
p=0.0378
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2